

1. Scalable Data Storage DONE

- Relational vs. NoSQL: Know when to use SQL vs. NoSQL databases. DONE
- Partitioning: Vertical and horizontal partitioning (sharding). Understand trade-offs. DONE
- Indexing: Covering indexes, primary vs. secondary indexes. DONE
- Consistency Models: Strong, eventual, causal DONE

Mongo DB scaling DONE

Postgres scaling

CAP

4. Asynchronous Processing

- Message Brokers: Kafka, RabbitMQ. When to use queues vs. streams.
- Event-Driven Architecture: Benefits of decoupling and event sourcing.
- Task Queues: For delayed jobs or retries.

DLQ

Kafka

9. API Design and Rate Management

- REST vs. GraphQL: Difference and practical use-cases for each.
- HATEOAS
- Pagination and Filtering: Strategies for efficiently fetching data.
 - API Versioning: Best practices for evolving APIs.
 - Throttle Requests: Why rate limiting is essential, algorithms like token bucket, leaky bucket.

2. Caching

- Client-side vs. Server-side Cache: Understand where caching should happen.
- Caching Strategies: Write-through, write-back, write-around.
- Distributed Cache: Redis, Memcached.
- Cache Eviction Policies: LRU, LFU, etc.

3. Load Balancing

- Horizontal Scaling: Why and how to horizontally scale services.
- Load Balancing Techniques: Round-robin, consistent hashing.
- Reverse Proxy: Understand how to use Nginx, HAProxy.

5. Database Read and Write Scaling

- Read Scaling: Master replication, read replicas.
- Write Scaling: Challenges with partitioning for writes, leader-election.
- CAP Theorem: Consistency, Availability, or Partition tolerance may be compromised.

6. Distributed Systems Concepts

- Consensus Algorithms: Paxos, Raft.
- Conflict Resolution: Last Write Wins, CRDTs, vector clocks for data reconciliation.

7. Reliability and Failover

- Redundancy: Active-passive vs. active-active configurations.
- Health Checks.
- Retries and Circuit Breakers: How to protect systems from cascading failures.

8. CDNs (Content Delivery Networks)

- Static Content Delivery: Why use a CDN, how does it work?
- Caching at the Edge: How CDNs improve latency for end users.

10. Search Systems

- Indexing: Building and maintaining indexes for fast search.
- Full-Text Search Engines: Elasticsearch, Azure AI Search.
- Ranking and Relevance: Basic understanding of how scoring works.

11. Monitoring and Observability and Security

- Metrics Collection: Prometheus, Grafana.
- Distributed Tracing: OpenTelemetry, Sentry.
- Centralized Logging.
- Authentication and Authorization: OAuth, JWT.
- Encryption: Data in transit vs. data at rest.

12. Disaster Recovery

RPO

RTO